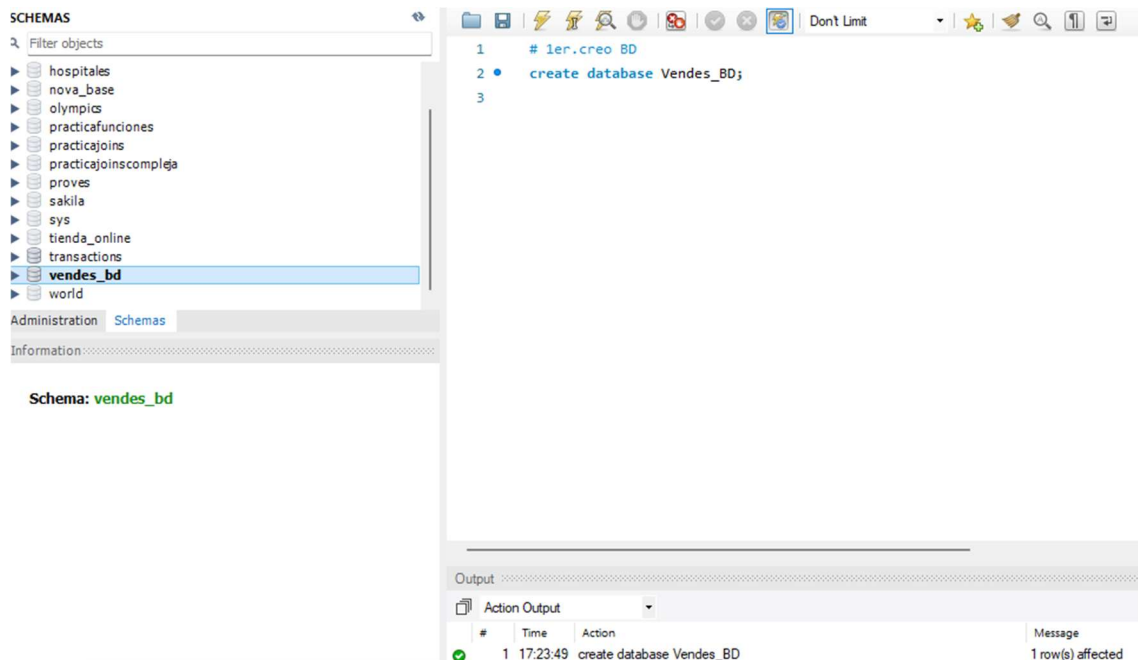


Partint d'alguns arxius CSV dissenyaràs i crearàs la teva base de dades.

Nivell 1

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

Creo la base de dades (BD) vendes_BD:



Vaig a crear les taules que formaran aquesta BD, inicialment dono una tipologia de dades de les columnes molt general i només determino les PK; Les Foreign Key les indicaré després perquè vull evitar el màxim els problemes d'importació de dades. També es per això que la tipologia de dades després l'adaptaré. Inicialment crearé 6 taules, però pel primer exercici només relacionaré 4 taules (companies, total_user (unió: european_user, american_user), credit_card, transactions després faré modificacions (taules auxiliars, arreglar les dades...):

Creo les taules de 'companies', 'europea_user', 'american_user', 'credit_card':

```
#creo la taula companies
create table if not exists companies (
    company_id varchar (100) not null primary key,
    company_name varchar (150),
    phone varchar (50),
    email VARCHAR(150),
    country VARCHAR(150),
    website VARCHAR(150)
);
show tables;
```

```

5 CREATE TABLE IF NOT EXISTS european_user (
7     id CHAR(10) PRIMARY KEY,
8     name VARCHAR(100),
9     surname VARCHAR(100),
10    phone VARCHAR(150),
11    email VARCHAR(150),
12    birth_date VARCHAR(100),
13    country VARCHAR(150),
14    city VARCHAR(150),
15    postal_code VARCHAR(100),
16    address VARCHAR(255)
17 );
18 # taula american_users
19 create TABLE IF NOT EXISTS american_user (
20     id CHAR(10) PRIMARY KEY,
21     name VARCHAR(100),
22     surname VARCHAR(100),
23     phone VARCHAR(150),
24     email VARCHAR(150),
25     birth_date VARCHAR(100),
26     country VARCHAR(150)
27 );

```

Output

#	Time	Action	Message
7	18:32:53	CREATE TABLE IF NOT EXISTS european_user (id CHAR(10) PRIMAR...	0 row(s) affected
8	18:33:47	create TABLE IF NOT EXISTS american_user (id CHAR(10) PRIMARY K...	0 row(s) affected

```

# crear taula credit_card
create table credit_card (
    id char (8) not null primary key,
    iban varchar (100) default null,
    pan varchar(25) default null,
    pin int default null,
    cvv int default null,
    track1 varchar (350) default null,
    track2 varchar (350) default null,
    expiring_date char(10) default null
);

```

Taula de productes:

```

52 #crear taula de products
53 create table products (
54     id int not null primary key,
55     product_name varchar (250) not null,
56     price decimal (10,2) not null,
57     colour varchar (50),
58     weight decimal (5,1),
59     warehouse_id varchar (15)
60 );
61
62
63

```

Output

#	Time	Action	Message
10	18:50:13	create table credit_card (id char (8) not null primary key, iban varchar (1...	0 row(s) affected
11	19:01:07	create table products (id int not null primary key, product_name varchar ...	0 row(s) affected

Taula de transactions:

Navigator

SCHEMAS

Filter objects

- Stored Procedures
- Functions
- vendes_bd
 - Tables
 - american_user
 - companies
 - credit_card
 - european_user
 - products
 - transactions
 - Views
 - Stored Procedures
 - Functions

Administration Schemas

Information

No object selected

```

63 • create table if not exists transactions(
64     id varchar (255) primary key,
65     card_id varchar (25),
66     business_id varchar (20),
67     timestamp timestamp,
68     amount decimal(10,2),
69     declined tinyint ,
70     product_ids varchar (25),
71     user_id int,
72     lat decimal (10,2),
73     longitude decimal (10,2)
74 );
75 • show tables;

```

Result Grid

Filter Rows:

Export: | Wrap Cell Content: I

Tables_in_vendes_bd	
american_user	
companies	
credit_card	
european_user	
products	
transactions	

Result 3 x

Output

Action Output

#	Time	Action	Message
14	19:23:07	create table if not exists transactions(idvarchar (255) primary key, card_j...	0 row(s) affected

Object Info Session

Importació de dades a taula Companies:

```

88
89 • LOAD DATA LOCAL INFILE 'C:/companies.csv'
90 INTO TABLE companies
91 CHARACTER SET utf8mb4
92 FIELDS TERMINATED BY ','
93 ESCAPED BY '\\'
94 IGNORE 1 ROWS
95 (company_id, company_name, phone, email, country, website);
96
97
98

```

Result Grid

Filter Rows:

Edit: | Export/Import: | Wrap Cell Content: I

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.comsite
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.comgroup9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.comsubcars
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.couk	Germany	https://cnn.comuser110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netflix.comsettings

companies 6 x

Output

Action Output

#	Time	Action	Message
12	13:17:27	select * from companies	100 row(s) returned
13	13:31:20	select * from companies	100 row(s) returned
14	13:31:22	select * from companies	100 row(s) returned

Importació de les dades de products:

```

113 • load data local infile 'C:/products.csv'
114 into table products
115 character set utf8mb4
116 FIELDS TERMINATED BY ','
117 IGNORE 1 ROWS
118 (id,product_name,@price,colour,weight,warehouse_id)
119 SET
120 id = id,
121 product_name = product_name,
122 price = REPLACE(@price, '$', '') + 0,
123 colour = colour,
124 weight = weight,
125 warehouse_id = warehouse_id;
126

```

Result Grid					
Filter Rows:					
Edit: Export/Import: Wrap Cell Content:					
id	product_name	price	colour	weight	warehouse_id
1	Direwolf Stannis	161.11	#7c7c7c	1.0	WH-4
2	Tarly Stark	9.24	#919191	2.0	WH-3
3	duel tourney Lannister	171.13	#d8d8d8	1.5	WH-2
4	warden south duel	71.89	#111111	3.0	WH-1
5	skywalker ewok	171.22	#bdbdbd	3.2	WH-0
6	dooku solo	136.60	#c4c4c4	0.8	WH--1

products 14 x

Output

Action Output

#	Time	Action	Message
46	15:09:04	load data local infile 'C:/products.csv' into table products character set utf8...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0

Taula american_users:

```

128
129 • load data local infile 'C:/american_users.csv'
130 into table american_user
131 character set utf8mb4
132 FIELDS TERMINATED BY ','
133 enclosed by '"'
134 IGNORE 1 ROWS
135 (id,name,surname,@phone,email,@birth_date,country,city,postal_code,@address)
136 SET
137 id = id,
138 name=name,
139 surname = surname,
140 phone= replace(replace(replace(replace(replace(@phone,'(',')',''),'-',''),'+',''),' ',''),
141 email= email,
142 birth_date= replace(@birth_date,' ',''),
143 country=country,
144 city=city,
145 postal_code=postal_code,
146 address= replace(@address,'#',' ')
147 ;

```

Output				
Action Output				
#	Time	Action	Message	Duration / I
56	17:15:35	load data local infile 'C:/american_users.csv' into table american_user char...		0.047 sec
57	17:16:45	select * from american_user	1010 row(s) returned	0.016 sec /

Comprovació:

165 • SHOW TABLES;
 166 • select *
 167 from american_user;
 168
 169
 170
 171
 172
 173
 174

id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	Zeus	Gamble	12825810551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.
10	Robert	McCarthy	3247466771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773
100	Melodie	Mclean	16772217152	risus.varius@google.ca	Sep 15, 1989	United States	San Jose	95101	Ap 644-8492 Sagittis St.
1006	Rbkocq	Swkmynuj	381868580	rbkocq.swkmynuj@example.com	Dec 2, 1989	United States	New York	10001	567 Swkmynuj St
101	Sarah	Beck	3586914345	vitae.risus@aol.co.uk	Apr 9, 1983	United States	Philadelphia	19101	665-9047 In
1011	Vosctr	Cpmnpofu	747268327	vosctr.cpmnpofu@example.com	Mar 21, 1954	United States	San Jose	95101	775 Cpmnpofu St
1015	Ydxylj	Yhswlpqn	847303250	ydxylj.yhswlpqn@example.com	Nov 13, 2000	Canada	Hamilton	L8E 0A1	209 Yhswlpqn St
1018	Unddlm	Znhkrinn	785371497	unddlm.znhkrinn@example.com	Jun 7, 1950	United States	Chicann	60601	47 Znhkrinn St

Output

#	Time	Action	Message	Duration / Fetch
62	17:21:32	load data local infile 'C:/american_users.csv' into table american_user char...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.047 sec
63	17:23:48	select * from american_user	1010 row(s) returned	0.000 sec / 0.000

Aquí veiem que hi ha una coma en els camps data; si es necessari ho arreglaré després, ahora de transformar les dades i adaptar-les.

Taula european_user:

```

191 • load data local infile 'C:/european_user.csv'
192 into table european_user
193 character set utf8mb4
194 FIELDS TERMINATED BY ','
195 enclosed by '"'
196 IGNORE 1 ROWS
197 (id,name,surname,@phone,email,@birth_date,country,city,postal_code,@address)
198 SET
199 id = id,
200 name=name,
201 surname = surname,
202 phone= replace(replace(replace(replace(replace(@phone,'(',')',''),'-',''),'+',''),' ',''),
203 email= email,
204 birth_date= replace(@birth_date,' ',''),
205 country=country,
206 city=city,

```

En 2 parts:

```

206 city=city,
207 postal_code=postal_code,
208 address= replace(@address,'#',' ')
209 ;
210

```

Comprovació:

```
227 • select *
228 from european_user;
229
230
231
232
233
234
235
```

Result Grid										
Filter Rows:										
Edit: Export/Import: Wrap Cell Content: Fetch rows:										
	id	name	surname	phone	email	birth_date	country	city	postal_code	address
	1137	Trmsqp	Ixwxjexx	955801451	trmsqp.ixwxjexx@example.com	Mar 23, 1970	Portugal	Amadora	2700-001	402 Ixwxjexx St
	1138	Euttga	Ncsgogdb	453236759	euttga.ncsgogdb@example.com	Feb 3, 2001	Germany	Cologne	50667	244 Ncsgogdb St
	1139	Hvxhyy	Erxspseu	466636706	hvxhyy.erxspseu@example.com	Sep 9, 1961	United Kingdom	Manchester	M1 1AE	673 Erxspseu St
	1140	Xwyloz	Bmhlqqey	503093843	xwyloz.bmhlqqey@example.com	Jan 14, 1964	Sweden	Malmö	211 43	352 Bmhlqqey St
	1142	Iuqoan	Ecgojqwh	528046097	iuqoan.ecgojqwh@example.com	Apr 18, 1980	Sweden	Örebro	703 61	409 Ecgojqwh St

european_user 4 x

Output

Action Output

#	Time	Action	Message
8	21:29:09	select * from european_user	3990 row(s) returned
9	21:34:55	select * from european_user	3990 row(s) returned

Creo a la taula credit_cards la columna user_id; per tant, ara afegeixo la columna:

Don't Limit										
215	#el posem darrera de id									
216 •	ALTER TABLE credit_card									
217	ADD COLUMN user_id INT									
218	AFTER id;									
219										
220										
221										
222										
223										
224										
225										

Result Grid										
Filter Rows:										
Edit: Export/Import: Wrap Cell Content:										
	id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date	
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

credit_card 10 x

Output

Action Output

#	Time	Action	Message
20	22:29:57	ALTER TABLE credit_card ADD COLUMN user_id INT AFTER id	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
21	22:30:07	select * from credit_card	0 row(s) returned

Dades de credit_card:

```

220 • load data local infile 'C:/credit_cards.csv'
221 into table credit_card
222 character set utf8mb4
223 FIELDS TERMINATED BY ','
224 enclosed by '"'
225 IGNORE 1 ROWS
226 (id,user_id,iban,pan,pin,card,track1,track2,expiring_date)
227 SET
228 id = id,
229 user_id=user_id,
230 iban = iban,
231 pan=pan,
232 pin=pin,
233 card = card,

```

Result Grid									
	id	user_id	iban	pan	pin	card	track1	track2	
	CcU-3050	259	FR5167744369175836831854477	4024007123722	4834	126	%B7784177223537930^NhuwoziEexiwa^9105...	%B2727867073512632=740369024475	
	CcU-3827	148	GB30AMTO46195520479336	491628 4555196727	3455	253	%B5272985556170790^CrxcdjjJpfvmr^32094...	%B4898288355167616=4309506673774	
	CcU-4485	54	GB30OKEB42532252608494	3443 989324 57829	6206	963	%B6777498848349615^VjkqhtjJvbqxv^11025...	%B2553542805713742=3704391407673	
	CcU-4758	15	GB51GUVH61469185263634	378486693428441	5241	571	%B2558446687136763^TdpmluFlanp^55114...	%B2852157356984972=360549723378	

```

224 enclosed by '"'
225 IGNORE 1 ROWS
226 (id,user_id,iban,pan,pin,card,track1,track2,expiring_date)
227 SET
228 id = id,
229 user_id=user_id,
230 iban = iban,
231 pan=pan,
232 pin=pin,
233 card = card,
234 track1=track1,
235 track2=track2,
236 expiring_date=expiring_date
237 ;

```

Result Grid									
	id	user_id	iban	pan	pin	card	track1	track2	
	CcU-3050	259	FR5167744369175836831854477	4024007123722	4834	126	%B7784177223537930^NhuwoziEexiwa^9105...	%B2727867073512632=740369024475	
	CcU-3827	148	GB30AMTO46195520479336	491628 4555196727	3455	253	%B5272985556170790^CrxcdjjJpfvmr^32094...	%B4898288355167616=4309506673774	
	CcU-4485	54	GB30OKEB42532252608494	3443 989324 57829	6206	963	%B6777498848349615^VjkqhtjJvbqxv^11025...	%B2553542805713742=3704391407673	
	CcU-4758	15	GB51GUVH61469185263634	378486693428441	5241	571	%B2558446687136763^TdpmluFlanp^55114...	%B2852157356984972=360549723378	

Carrego les dades de la taula transactions:

```

238  ## cargo les dades de transactions
239  • load data local infile 'C:/transactions.csv'
240  into table transactions
241  character set utf8mb4
242  FIELDS TERMINATED BY ';'
243  IGNORE 1 ROWS
244  (id,card_id,business_id,@timestamp,amount,declined,product_ids,user_id,lat,longitude)
245  SET
246  timestamp = STR_TO_DATE(@timestamp, '%Y-%m-%d %H:%i:%s')
247  ;

```

Hem dóna errors perquè les taules lat i longitude les tinc definides com a decimals, haig de canviar-les:

```

}    ### faig que siguin varchar
} • ALTER TABLE transactions
1    MODIFY lat VARCHAR(20),
2    MODIFY longitude VARCHAR(20);
}

```

I ara torno a carregar les dades -previ el borrar les dades de la importació amb errors anteriors-:

```

253
254  ## torno a cargo les dades de transactions
255  • load data local infile 'C:/transactions.csv'
256  into table transactions
257  character set utf8mb4
258  FIELDS TERMINATED BY ';'
259  IGNORE 1 ROWS
260  (id,card_id,business_id,@timestamp,amount,declined,product_ids,user_id,lat,longitude)
261  SET
262  timestamp = STR_TO_DATE(@timestamp, '%Y-%m-%d %H:%i:%s')
263  ;
264

```

id	card_id	business_id	timestamp	amount	declined	product_ids	user_id	lat	longitude
00043A49-2949-494B-A5DD-A5BAE38B19DD	Cc5-9294	b-2458	2024-08-28 07:16:46	395.43	0	16, 26, 97, 87	4713	46.19992926158272	1.4355402821327607
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	Cc5-5019	b-2370	2016-12-21 20:07:18	155.63	0	66, 69, 87	438	41.59720554463741	12.22175994259365
00045D68-ED2E-4F2F-8186-CEE074D875D0	Cc5-6699	b-2390	2020-07-14 15:37:45	326.01	0	30, 11, 16, 81	2118	29.757295899964348	-95.37963676492454
000481C3-1C26-4FEF-83A0-4CD0EB0046BD	Cc5-6696	b-2230	2017-09-04 19:44:53	161.60	0	72	2115	53.54888376797025	-113.50305274646564
00051AA4-9CBE-4268-B070-C38062A1B3E2	Cc5-7606	b-2266	2017-01-05 18:19:25	148.91	0	18	3025	52.20836951654172	5.690806474241335

transactions 28 x

#	Time	Action	Message	Duration
85	19:28:40	select * from transactions	0 row(s) returned	0.000 s
86	19:29:58	load data local infile 'C:/transactions.csv' into table transactions character set utf8mb4 FI...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0	2.344 s
87	19:30:06	select * from transactions	100000 row(s) returned	0.000 s

Ara aniré fent modificacions i canvis en les taules per tenir un format més arreglat, FK... vaig mirant taula a taula i fent canvis i comentant:

Comprovo si hi ha alguna taula amb columnes null, per exemple faig:

```

271  • SELECT *
272  FROM companies
273  WHERE company_id IS NULL OR company_name IS NULL
274  OR phone IS NULL OR email IS NULL OR country IS NULL OR website IS NULL;

```

Ho miro en totes les taules.

```

select*
from credit_card;
SELECT *
FROM credit_card
WHERE id IS NULL OR user_id IS NULL
OR iban IS NULL OR pan IS NULL OR pin IS NULL OR cvv IS NULL OR track1 is null OR track2 is null OR expiring_date is null;

```

Canvio nom de les columnes per tenir-les iguals en taula de fets i de dimensions per facilitar la lectura, en aquest cas passo a la columna business_id de la taula transactions a company_id – nom que rep a la taula companies on es PK, aprofito per posa-li el mateix varchar.

```

276  ## canvio el nom en transaccions per posar el mateix nom a taules dimensions i types
277  • alter table transactions
278    change business_id company_id varchar (100);
279  ### comprovacio
280  • select*
281    from transactions;
282
283  ### la converteixo en FK
284  • alter table transactions
285    add constraint fk_transactions_companies
286    foreign key (company_id)
287    references companies (company_id)
288    on update cascade
289    on delete cascade;
290
291
292
293
294

```

Output :

#	Time	Action	Message
28	11:21:45	alter table transactions change business_id company_id varchar (100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
29	11:23:37	select* from transactions;	100000 row(s) returned
30	11:24:06	alter table transactions add constraint fk_transactions_companies foreign key (company_id) references companies (company_id) on update cascade on delete cascade;	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0

En taula de credit_card, faig canvi de nom, determino la relació (FK de taula transactions)...

```

289  #### canvis en Product
290  ## canvio el nom en credit_card per posar el mateix nom a taules dimensions i types
291  • alter table credit_card
292    change id card_id varchar (25);
293  ### comprovació
294  • select*
295    from credit_card;
296
297
298
299

```

Result Grid

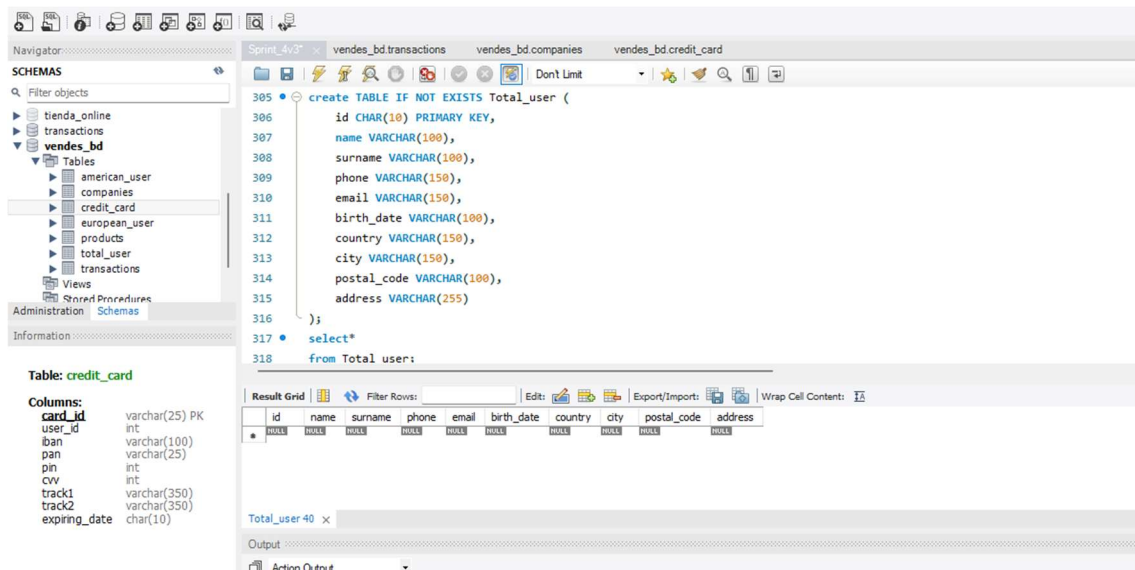
card_id	user_id	iban	pan	pin	cvv	track1	track2	e
CcS-4857	276	XX4857591835292505850771	2314242385113924	1819	467	%B2314242385113924^LWCBUDLWCBUD^22...	%B2314242385113924=2410101518363164?	0?
CcS-4858	277	XX8581768137002436094025	6582720299715533	3964	817	%B6582720299715533^TIQMVITIQMVI^2404...	%B6582720299715533=2411101104546272?	1?
CcS-4859	278	XX7826930491423553609370	8861684536289642	4983	277	%B8861684536289642^COFBGDCOFBGD^280...	%B8861684536289642=2502101761665371?	1?
CcS-4860	279	XX5559590368835304645299	2481155515498459	6876	661	%B2481155515498459^TIUJTUTUJTU^31040...	%B2481155515498459=26021015144414395?	0?

credit card 29 x

Output :

#	Time	Action	Message	Duration
38	11:47:06	select* from credit_card	5000 row(s) returned	0.000 s
39	11:47:11	alter table credit_card change id card_id varchar (25)	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.250 s
40	11:47:16	select* from credit_card	5000 row(s) returned	0.016 s

El veure que hi ha 2 taules user i una estructura similar, he decidit fusionar-la (unir-los) amb una sola:



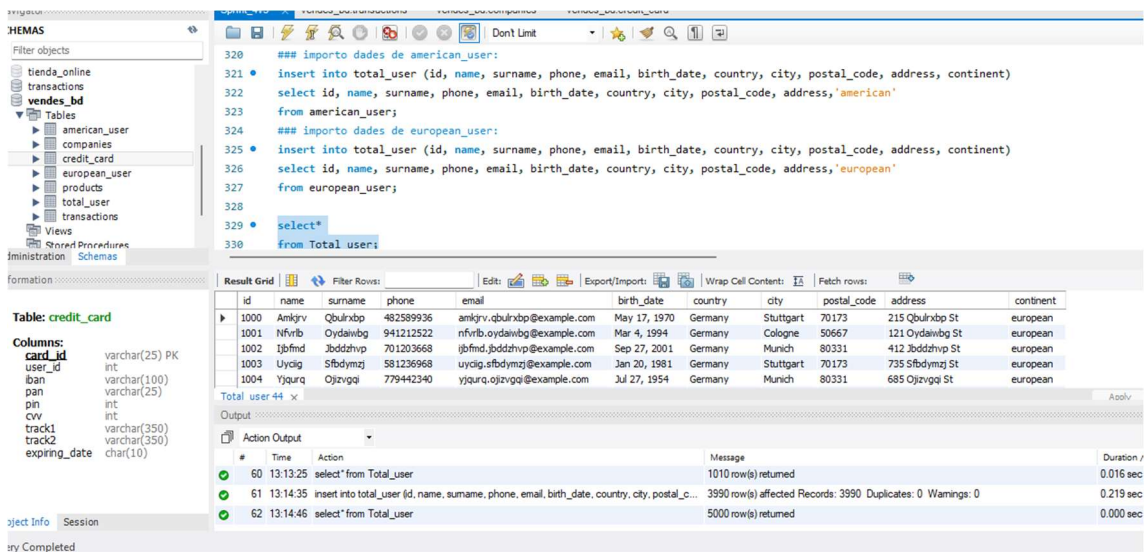
Afegim columna continent:

```

7   ### afegieixo columna continent
8 •   alter table total_user
9   add continent varchar (50) not null;

```

Importem dades dels 2 continents i comprovem nova taula Total_user:



Anem per fer la FK, però amb canvis previs:

```

329  ## canvio el nom en credit_card per posar el mateix nom a taules dimensions i types
330  • alter table total_user
331  change id user_id int;
332  ### comprovació
333  • select*
334  from total_user;
335
336
337
338
339

```

	user_id	name	surname	phone	email	birth_date	country	city	postal_code	address	continent
1	Zeus	Gamble		12825810551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.	american
2	Garrett	Mcconnell		7182572412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave	american
3	Ciaran	Harrison		5225981365	interdum.feugiat@aol.org	Apr 29, 1998	United States	Houston	77001	736-2063 Tellus St.	american
4	Howard	Stafford		14117403269	ornare.egestas@idoud.edu	Feb 18, 1989	United States	Phoenix	85001	Ap 545-2244 Erat. Rd.	american
5	Hayfa	Pierce		15545412077	et.malesuada.fames@hotmail.org	Sep 26, 1998	United States	Philadelphia	19101	341-2821 Ultrices Av.	american

total user 47 x

Output

#	Time	Action	Message
64	13:26:59	select* from total_user	5000 row(s) returned
65	13:27:05	alter table total_user change id user_id int	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
66	13:27:11	select* from total_user	5000 row(s) returned

FK:

```

335  ### determino fk en transactions vs Total_user
336  • alter table transactions
337  add constraint fk_transactions_user_id
338  foreign key (user_id)
339  references total_user (user_id)
340  on update cascade
341  on delete cascade;
342
343
344
345
346
347
348
349
350
351
352

```

Output

#	Time	Action	Message
65	13:27:05	alter table total_user change id user_id int	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
66	13:27:11	select* from total_user	5000 row(s) returned
67	13:30:59	alter table transactions add constraint fk_transactions_user_id foreign key (user_id) refer...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0

Ara ja podem borra les taules american_user i european_user:

```

385
386     ##### Eliminem taules sobrants:
387 •   drop table american_user;
388 •   drop table european_user;
389     ### comprovacio
390
391 •   show tables;
392
393
394
395

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Tables_in_vendes_bd

- companies
- credit_card
- products
- taula_products
- total_user
- transactions

Result 35 x

Output

Action Output

#	Time	Action	Message
54	21:30:47	drop table european_user	0 row(s) affected
55	21:31:36	show tables	6 row(s) returned

Exercici 1

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```

343     ###Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.
344 •   select*
345     from total_user
346     join ( select t.user_id, count(t.id) as Num_trans
347           from transactions t
348           group by t.user_id) as sub_tabla
349     on sub_tabla.user_id=total_user.user_id
350     where sub_tabla.Num_trans > 80;
351
352
353

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

user_id	name	surname	phone	email	birth_date	country	city	postal_code	address	continent	user_id	Num_b
185	Molly	Gilliam	08001208023	donec@outlook.couk	Dec 21, 1993	United Kingdom	London	EC1A 1BB	P.O. Box 202, 5638 Mi Rd.	european	185	110
289	Dxwgi	Hwcru	983098797	dxwgi.hwcru@example.com	Aug 20, 1976	Germany	Stuttgart	70173	82 Hwcru Street	european	289	94
318	Bnyr	Astuw	331209644	bnyr.astuw@example.com	May 3, 1974	Italy	Genoa	16100	53 Astuw Street	european	318	91
454	Sfzoh	Xgvfridx	584953945	sfzoh.xgvfridx@example.com	Aug 28, 1962	Poland	Gdansk	80-001	52 Xgvfridx Street	european	454	81

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetc
7	10:30:23	select* from total_user join (select t.user_id, count(t.id) as Num_trans from transactions t ...	4 row(s) returned	0.469 sec / 0.0
8	10:33:31	select* from total_user join (select t.user_id, count(t.id) as Num_trans from transactions t ...	4 row(s) returned	0.360 sec / 0.0

Exercici 2

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

357 **###Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.**
 358 • **select cc.iban, avg(t.amount) as Mitj_iban**
 359 **from credit_card cc**
 360 **join transactions t**
 361 **on cc.card_id=t.card_id**
 362 **join companies c**
 363 **on t.company_id=c.company_id**
 364 **where c.company_name='Donec Ltd'**
 365 **group by cc.iban;**
 366
 367
 368

Result Grid | Filter Rows: | Exports: | Wrap Cell Content:

iban	Mitj_iban
XX911406401125586307586805	356.246667
SK9446370242474562577506	142.960000
XX776752917845952975555640	257.370000
XX413827362289719304908990	139.590000
XX347787246070769610780308	240.410000

Result 23 x

Output

Action Output

#	Time	Action	Message
47	13:17:47	select cc.iban, avg(t.amount) from credit_card cc join transactions t on cc.card_id=t.card...	371 row(s) returned
48	13:18:17	select cc.iban, avg(t.amount) as Mitj_iban from credit_card cc join transactions t on cc.c...	371 row(s) returned

Nivell 2

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en: si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:...

Creo la taula:

544
 545 **###Creem la taula**
 546 • **CREATE TABLE estat_targeta (**
 547 **id INT AUTO_INCREMENT PRIMARY KEY,**
 548 **card_id VARCHAR(25),**
 549 **tt VARCHAR(10),**
 550 **FOREIGN KEY (card_id) REFERENCES credit_card(card_id)**
 551 **);**
 552
 553
 554
 555
 556
 557 **###Inserim les dades calculades**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	card_id	tt
1	CcS-4870	inactiva
2	CcS-4899	inactiva
3	CcS-4904	activa
4	CcS-4936	activa
5	CcS-4942	activa
6	CcS-4908	inactiva

Estat_targeta 2 x

Output

Action Output

#	Time	Action	Message
5	19:08:57	CREATE TABLE estat_targeta (id INT AUTO INCREMENT PRIMARY KEY, card...	0 row(s) affected

Creem CTE ens permet assignar un nom a un conjunt de resultats per després poder-los utilitzar amb altres operacions:

```

549
550 #creo la CTE -per tenir cada targeta si ha estat declined o amb una sequencia entera
551 ##assignar un nom a un conj.de resultats temporals
552 • with targeta as (
553     select
554         transactions.card_id,
555         transactions.declined,
556         row_number ()over( partition by transactions.card_id order by transactions.timestamp desc) as ttt
557     from transactions
558 )
559 select *
560 from targeta;

```

card_id	declined	ttt
CcS-4857	0	1
CcS-4857	0	2
CcS-4857	0	3
CcS-4857	1	4
CcS-4857	0	5

Result 2 x

Output

#	Time	Action	Message
1	10:56:56	select tp.products_ids, p.product_name, count(tp.products_ids) as num_vend from taula_p...	100 row(s) returned
2	11:24:33	with targeta as (select transactions.card_id, transactions.declined, row_numbe...	100000 row(s) returned

```

102 • INSERT INTO estat_targeta (card_id, tt)
103 WITH targeta AS (
104     SELECT
105         transactions.card_id,
106         transactions.declined,
107         transactions.id,
108         ROW_NUMBER() OVER (
109             PARTITION BY transactions.card_id
110             ORDER BY transactions.timestamp DESC
111         ) AS ttt
112     FROM transactions
113 )
114 SELECT
115     card_id,
116     CASE
117         WHEN SUM(CASE WHEN declined = 1 THEN 1 ELSE 0 END) = 3
118             AND COUNT(ttt) = 3
119     THEN 'inactiva'

```

Output

#	Time	Action	Message
37	13:28:08	with targeta as (select transactions.card_id, transactions.declined, row_numbe...	100000 row(s) returned
38	13:28:54	INSERT INTO estat_targeta (card_id, estat) SELECT cc.id, CASE WHEN SU...	Error Code: 1146. Table 'vendes_bd.targeta' doesn't exist
39	13:32:18	INSERT INTO estat_targeta (card_id, tt) WITH targeta AS (SELECT transactions.c...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0

En 2 parts:

```

SELECT
    card_id,
    CASE
        WHEN SUM(CASE WHEN declined = 1 THEN 1 ELSE 0 END) = 3
            AND COUNT(ttt) = 3
        THEN 'inactiva'
        ELSE 'activa'
    END AS tt
FROM targeta
WHERE ttt <= 3
GROUP BY card_id;

```

Exercici 1

Quantes targetes estan actives?

```
388  ###Exercici 1
389  ###Quantes targetes estan actives?
390
391  • SELECT COUNT(*) AS targetes_actives
392  FROM estat_targeta
393  WHERE tt = 'activa';
394
395
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

targetes_actives
4995

Result 2 x

Output

Action Output

#	Time	Action	Message
40	13:39:28	SELECT COUNT(*) AS targetes_actives FROM estat_targeta WHERE tt = 'activa'	1 row(s) returned
41	13:39:45	create database Vendes_BD	Error Code: 1007. Can't
42	13:39:53	SELECT COUNT(*) AS targetes_actives FROM estat_targeta WHERE tt = 'activa'	1 row(s) returned

Nivell 3

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

Abans de crear la relació amb productes, tenim que “desnormalitzar”(Json_table) la columna product_ids . Alerta tenim dades en “;”. En aquest cas creo una taula amb totes les columnes i després ja treure les innecessàries.

```
create table taula_products as
select
  tr.id,
  tr.card_id,
  tr.company_id,
  tr.timestamp,
  tr.amount,
  tr.declined,
  jtable.products_ids,
  tr.user_id,
  tr.lat,
  tr.longitude
from transactions tr,
json_table(
  concat('[',replace (tr.product_ids,';',',',''),']'),
  '$[*]' columns(
    products_ids varchar(10) path '$'
  )
) as jtable;
```

Arreglo les dades:

```

380     ### creo PK de taula_products
381 •   alter table taula_products
382     add primary key (id,products_ids);
383     ### FK taula transaction
384 •   alter table taula_products
385     add constraint fk_transaction
386     foreign key (id)
387     references transactions(id)
388     on update cascade
389     on delete cascade;
390     ###canviar el tipus de products_ids a INT
391 •   ALTER TABLE taula_products
392     MODIFY COLUMN products_ids INT;
393     ### FK taula products
394 •   alter table taula_products
395     add constraint fk_products_ids
396     foreign key (products_ids)
397     references products(id)
398     on update cascade
399     on delete cascade;

```

Em quedo amb les columnes necessàries per la taula:

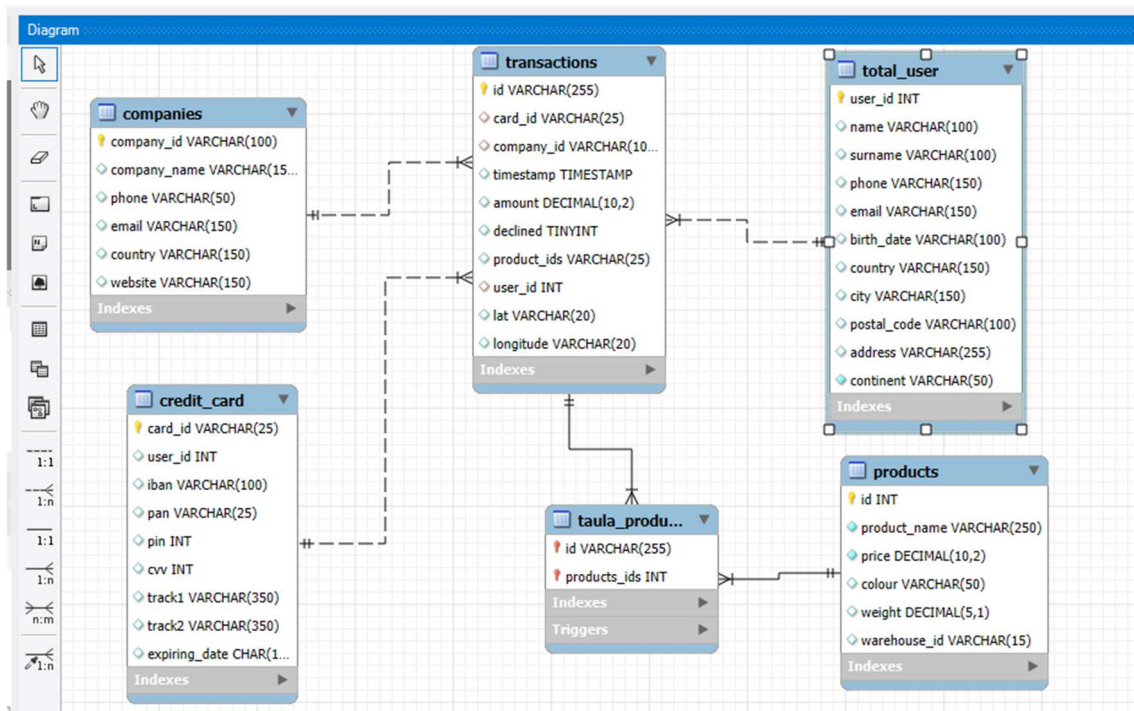
```

#### eliminar columnas de taula_products sobrants a Json_table
•   alter table taula_products
    drop card_id,
    drop company_id,
    drop timestamp,
    drop amount,
    drop declined,
    drop user_id,
    drop lat,
    drop longitude;

•   select*
    from taula_products;

```

Diagrama final:



Exercici 1

Necessites conèixer el nombre de vegades que s'ha venut cada producte.

```
454 #Exercici 1
455 #Necessitem conèixer el nombre de vegades que s'ha venut cada producte. declined=0
456 select tp.products_ids, p.product_name, count(tp.products_ids) as num_vend
457 from taula_products tp
458 join products p
459 on tp.products_ids=p.id
460 join transactions t
461 on tp.id=t.id
462 where t.declined = 0
463 group by tp.products_ids,p.product_name
464 order by num_vend desc;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	products_ids	product_name	num_vend
▶	52	riverlands the duel	2642
	29	Tully maester Tarly	2627
	21	duel Direwolf	2603
	16	the duel warden	2602
	33	duel warden	2593
	27	with trade	2501

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 58	13:41:25	select* from products	100 row(s) returned